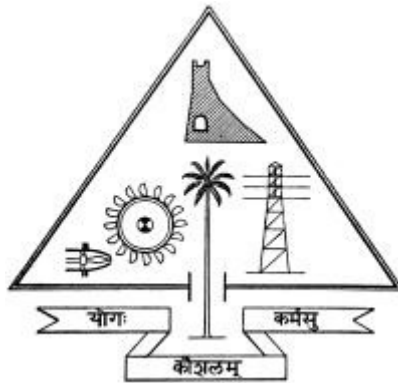


LATEX WRITESENSE AI

*Main Project Report submitted in partial fulfillment of the requirements for the
award of the degree of **Master of Computer Applications** of the **APJ Abdul
Kalam Technological University***

submitted by

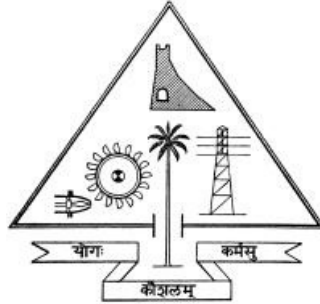
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CERTIFICATE

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DECLARATION

I hereby declare that the main project named, **LATEX WRITESENSE AI**, is my own work and that, to the best of my knowledge and belief, it contains no material previously published by another person nor material which has been accepted for the award of any other degree or course of the university or any other institute of higher learning, except where due acknowledgment and reference has been made in the text.

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ABSTRACT

Writing in LaTeX environments such as Overleaf usually involves a lot of revision, especially when trying to improve grammar, clarity, and overall readability. However, most AI writing tools are not built directly into the LaTeX workflow. Because of this, users often end up switching between different platforms to edit and refine their work. This can interrupt the writing process and make it harder to stay focused. In some cases, it may also lead to small changes in meaning or even cause issues with LaTeX formatting. Another concern is the amount of control users have over these tools. Fully automatic corrections might seem convenient, but they can sometimes make changes that are not very transparent. As a result, users may find it difficult to understand or trust what has been modified in their writing. To deal with these issues, this work presents a lightweight and context-aware AI writing support system designed specifically for LaTeX documents. Instead of working on the entire document at once, the system focuses only on the specific sentences or paragraphs selected by the user. It provides several helpful features such as grammar correction, improving readability, summarizing content, translating text into different languages, and even suggesting relevant research papers. Rather than relying on full document analysis, the system uses task-specific instructions to understand the context. This helps in making accurate improvements while still keeping the original meaning and maintaining an academic tone. It also includes LaTeX-aware editing, which ensures that important elements like commands, citations, and equations are not accidentally changed during processing. By bringing controlled AI support directly into the LaTeX environment, the system aims to improve writing quality without taking away user control. The design

is simple enough for student-level projects, but still effective in solving common writing challenges. It also allows users to make small, step-by-step improvements instead of changing everything at once. This way, the overall structure of the document remains intact while the quality of writing gradually improves. In the end, this approach creates a better balance between human effort and AI assistance, resulting in clearer and more polished academic writing.

CONTENTS

List of Figures	vii
Nomenclature	ix
1 INTRODUCTION	1
2 LITERATURE REVIEW	4
3 ENVIRONMENTAL STUDY	8
3.1 System Configuration	9
3.1.1 Hardware Requirements	9
3.1.2 Software Requirements	10
3.1.3 Software Requirements for Development	11
3.2 Software Specification	12
3.2.1 Python 3.9	12
3.2.2 FastAPI	13
3.2.3 LanguageTool	13
3.2.4 Large Language Models and Transformer Models	13
3.2.5 CrossRef API	14
3.2.6 JavaScript and Chrome Extension APIs	14
3.2.7 JSON-Based Structured Communication	14
4 SYSTEM ANALYSIS	16
4.1 Requirements Analysis	16
4.2 Existing System	17
4.3 Limitations of Existing System	17
4.4 Proposed System	18
4.5 Advantages of Proposed System	19
4.6 Functional Requirements	20
4.6.1 User Interface	20
4.6.2 User Input	20
4.6.3 Grammar Correction	21
4.6.4 Clarity Analysis	21
4.6.5 Summarization of Selected Text	21
4.6.6 Multilingual Translation	21
4.6.7 Related Research Paper Identification	21
4.6.8 Apply and Control Mechanism	22
4.7 Non-Functional Requirements	22
4.7.1 Performance	22
4.7.2 Usability	23
4.7.3 Compatibility	23
4.7.4 Scalability	23
4.7.5 Reliability	24
4.7.6 Security	24

4.8	Feasibility Study	24
4.8.1	Technical Feasibility	24
4.8.2	Operational Feasibility	25
4.8.3	Economic Feasibility	25
5	SYSTEM DESIGN	26
5.1	System Architecture	26
5.2	Data Flow Diagram	27
5.2.1	DFD Level 0	28
5.2.2	DFD Level 1	29
6	SYSTEM IMPLEMENTATION	31
6.1	Development Environment Setup	32
6.2	Chrome Extension Implementation	33
6.3	Grammar Correction Engine	33
6.3.1	Rule-Based Grammar Correction	34
6.3.2	AI-Based Structural Grammar Improvement	34
6.4	Academic Clarity Analysis Engine	35
6.4.1	Readability Metrics	35
6.4.2	Structural Academic Metrics	35
6.5	Text Summarization Module	36
6.6	Multilingual Translation Module	37
6.7	Research Paper Identification Module	37
6.8	System Integration	37
7	RESULTS AND DISCUSSION	39
7.1	System Interface Overview	39
7.2	Grammar Correction Results	41
7.3	Clarity Evaluation Results	42
7.4	Text Summarization Results	44
7.5	Multilingual Translation Results	45
7.6	Research Paper Identification Results	47
8	CONCLUSION	51
9	FUTURE ENHANCEMENT	52
	BIBLIOGRAPHY	53

LIST OF FIGURES

5.1	Architecture Diagram	27
5.2	Data Flow Diagram Level 0	28
5.3	Data Flow Diagram Level 1	29
7.1	WriteSense AI Extension Interface in Overleaf	40
7.2	Grammar Correction Module Output	42
7.3	Clarity Analysis Module Output	43
7.4	Text Summarization Output	44
7.5	Multilingual Translation Output	46
7.6	Research Paper Identification Module Output	49

NOMENCLATURE

AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
FKG	Flesch–Kincaid Grade Level
FRE	Flesch Reading Ease
JSON	JavaScript Object Notation
LLaMA	Large Language Model Meta AI
LLM	Large Language Model
NLLB	No Language Left Behind
NLP	Natural Language Processing
REST	Representational State Transfer

CHAPTER 1

INTRODUCTION

Good and effective academic writing is a vital aspect for communicating research contributions, especially in technical and scientific domains. Research papers, theses, and technical reports are expected to convey complex ideas in a logical and academically appropriate manner. However, achieving high-quality writing is not just limited to grammatical accuracy but also involves clarity, logical flow, and compliance with academically appropriate writing styles. However, achieving these aspects can be a challenge for authors, especially for writing long documents.

LaTeX-based authoring platforms such as Overleaf are highly popular in academic communities for authoring documents. This is mainly due to their robust capabilities for authoring structured documents. These tools are commonly used by authors in various technical domains such as computer science, engineering, mathematics, and physical sciences. However, these platforms are mainly focused on document authoring and have poor support for improving linguistic quality. Grammar correction, readability improvement, and clarity improvement are mostly manual processes. However, many authors prefer to use third-party grammar-checking tools or AI-based writing assistants for improving linguistic quality. However, these tools are mostly independent applications and have poor integration with LaTeX-based authoring platforms. This creates inefficient authoring workflows for users, as authors are forced to switch between tools frequently. Moreover, there are high chances of document inconsistencies while copying data between applications. Moreover, processing LaTeX code outside the LaTeX environment may change LaTeX commands or citations

or math expressions, resulting in errors during document compilation.

This has led to significant developments in writing assistants based on artificial intelligence and natural language processing. These writing assistants are now capable of identifying grammatical errors, assessing clarity, evaluating readability, and even summarizing text. These developments have opened new opportunities for artificial intelligence integration in academic writing. However, most writing assistants are based on completely automatic rewriting. This may compromise transparency and control for the author over their writing.

To address the challenges faced by authors during academic writing, this project introduces an AI-assisted writing support tool called WriteSense AI. The tool is specifically designed for LaTeX-based platforms such as Overleaf. It is implemented as a Chrome extension, allowing it to function directly within the main writing interface. This integration helps avoid unnecessary distractions, enabling authors to focus on their work without switching between different tools.

WriteSense AI is developed using concepts from artificial intelligence and natural language processing, with the aim of supporting the writing process rather than replacing it. One of its key features is that it operates only on user-selected text. This gives authors better control over their content, as they can choose exactly which part of the text needs improvement. Based on the selected portion, the system can perform various tasks such as grammar correction, clarity analysis, translation, summarization, and even suggesting relevant research papers.

The system follows a modular client-server architecture. The Chrome extension acts as the front-end, where the user interacts with the tool. It captures the selected text and sends it to backend services developed using the FastAPI framework. Each feature is handled as an independent module, making the system easier to manage and extend. For example, grammar correction is carried out using rule-based linguistic methods and AI-based language models, ensuring both accuracy and contextual relevance in the

generated suggestions while maintaining scalability of the system. An important aspect of the system is its ability to handle LaTeX content properly. Since LaTeX documents include commands, equations, and citation formats, it is essential that these elements are not altered during processing. To ensure this, the system uses preprocessing techniques to identify and preserve LaTeX-specific components before applying any modifications.

Apart from improving writing quality, the system also supports literature exploration. By analyzing key terms from the selected text, it can suggest relevant research papers from academic databases. These recommendations are made available within the system itself, making it easier for users to explore related work without leaving the writing environment.

In summary, WriteSense AI aims to make the academic writing process more efficient and user-friendly by integrating AI support directly into the LaTeX workflow. To provide a strong foundation for the proposed system, the following chapter presents a comprehensive literature review, analyzing existing research works, methodologies, and advancements in AI-based academic writing assistance systems.

CHAPTER 2

LITERATURE REVIEW

Over the last few years, the impact of artificial intelligence on the conduct of research and writing has increased significantly. Writing in research is not just about conveying ideas; it is about being clear, well-organized, and precise. Moreover, the amount of research being published is growing exponentially, which is causing challenges for researchers in effectively handling, understanding, and communicating research information. This highlights the limitations of traditional writing approaches, where tasks such as grammatical correction, structural improvement, and literature review must be performed separately, making the process time-consuming.

Academic writing is considered an essential part of research, especially in technical and scientific domains. While writing research papers, authors are expected to present ideas in a logical sequence, use consistent terminology, and follow standard academic conventions. However, achieving this level of quality is not always easy. Many students and early-stage researchers struggle with maintaining clarity, grammatical correctness, and an appropriate academic tone. This has led to increasing interest in intelligent systems that assist users during the writing process.

The use of artificial intelligence in academic writing has become increasingly important, as techniques such as large language models, transformer architectures, and machine learning algorithms enable advanced text analysis. These technologies can be used for grammar correction, readability assessment, summarization, and content improvement. As a result, AI-based writing support systems have emerged as a significant research area.

Hou et al. [1] developed a multi-agent system called PaperDebugger, which integrates AI tools directly into LaTeX-based writing environments. Unlike traditional tools, this system operates within the editor, allowing continuous interaction without disrupting the writing workflow. The framework consists of multiple agents responsible for different tasks such as text evaluation and content refinement. This approach improves efficiency by enabling simultaneous processing of multiple writing tasks while preserving document structure.

Totawad et al. [2] proposed a research paper recommendation system based on deep learning and large language models. The system focuses on understanding semantic relationships within research documents using contextual representations. This enables the identification of relevant literature and research trends. Such systems are highly beneficial in academic writing, where identifying related work is a critical component.

Swathi et al. [3] investigated machine learning-based techniques for automatic detection of syntactic and semantic errors in English writing. Unlike traditional rule-based systems, their approach analyzes relationships between words and sentence structures to identify issues such as incorrect tense, awkward phrasing, and contextual inconsistencies. This demonstrates that machine learning methods can significantly enhance grammar correction systems.

Jovic et al. [4] explored the use of large language models for providing automated feedback in academic writing. The study compared AI-generated feedback with human evaluation and found that AI systems offer fast and consistent suggestions. However, human reviewers provide deeper contextual understanding. This highlights the importance of combining AI efficiency with human insight in writing systems.

Ilango [5] proposed a deep learning-based abstractive text summarization model using a sequence-to-sequence architecture. The model uses an encoder-decoder framework with attention mechanisms to generate meaningful summaries while preserving contextual information.

This approach demonstrates the effectiveness of NLP techniques in reducing the complexity of large textual content, which is highly relevant for academic writing systems.

Das and Das [6] studied the role of Natural Language Processing in improving human-computer interaction. The research highlights that advancements in AI and deep learning have significantly enhanced the ability of machines to understand and generate human language. NLP techniques enable better interaction, opinion analysis, and decision-making support, which are essential for intelligent writing assistance systems.

Challener et al. [7] analyzed the use of readability metrics, particularly the Flesch-Kincaid Grade Level, to evaluate the clarity of textual content. Their study demonstrated that readability scores can effectively measure the complexity and comprehensibility of documents. This highlights the importance of incorporating readability analysis in writing systems to ensure that generated text is accessible and easy to understand.

Lee et al. [8] presented a deep learning-based approach for context-sensitive spelling error correction. The system utilizes various language models, including word embeddings and transformer-based architectures, to detect and correct errors based on contextual understanding. The study shows that autoencoding models such as BERT and RoBERTa achieve high accuracy by leveraging bidirectional context, significantly outperforming traditional methods.

The reviewed studies demonstrate that AI-based approaches significantly enhance academic writing by improving grammar correction, readability assessment, summarization, and semantic analysis. These techniques enable automated text refinement, reduce manual effort, and improve overall writing quality. Furthermore, the integration of machine learning and deep learning models allows systems to understand contextual meaning and generate more accurate suggestions. Such advancements highlight the growing importance of intelligent writing support systems in research environments.

The proposed WriteSense AI system builds upon these advancements by integrating multiple AI-based modules into a unified platform. It combines grammar correction, clarity evaluation, summarization, translation, and research assistance to provide comprehensive writing support. This integration improves efficiency by offering multiple functionalities within a single interface. As a result, the system enhances both productivity and writing quality. The next chapter presents the system design and implementation of the proposed WriteSense AI system. It describes the overall architecture, key modules, and technologies used to develop the system. The chapter also explains how different components such as grammar correction, readability analysis, summarization, translation, and research assistance are integrated into a unified platform. It outlines the interaction between the frontend and backend, along with the tools and frameworks used for development.

CHAPTER 3

ENVIRONMENTAL STUDY

This chapter provides an overview of the hardware and software environment necessary for the development and execution of the proposed WriteSense AI system. This includes the necessary hardware requirements, software tools, frameworks, and libraries necessary for the execution of the proposed AI system for grammar correction, clarity analysis, research paper identification, summarization, and translation of text into different languages using the LaTeX editor.

The proposed AI system is composed of a Chrome browser extension and several AI-powered services implemented using the FastAPI framework. System configuration is necessary for the execution of the proposed AI system. This is important for the stable execution of the browser extension, backend processing, AI model inference, and interaction with external databases.

The proposed AI system is developed using a combination of different technologies that are required for its proper working. All these components need to be connected and work together smoothly so that the system runs without any major issues. This chapter mainly discusses the environment needed to run the system in a practical situation. It also explains the basic hardware components required to maintain good performance.

Along with that, the chapter includes the software requirements such as the frameworks, tools, and environments used during development and execution. Some additional aspects like system compatibility, configuration, and resource usage are also considered to make sure the system works in a stable and efficient way.

3.1 System Configuration

System configuration is important because it helps in identifying the hardware setup required to run the proposed system. It gives a clear idea about what kind of system is needed for proper execution.

The system is designed to support transformer-based natural language processing models for tasks like summarization and translation. Since these models process large amounts of data and depend on context, they need enough memory and processing power. Also, a proper internet connection is required because the system depends on online platforms like Overleaf and external research databases.

3.1.1 Hardware Requirements

The hardware requirements define the minimum system specifications needed to use the application properly. Since the system performs multiple NLP tasks and interacts with AI models, it requires decent computing resources.

- **Memory:** At least 8 GB RAM is required for smooth working of the backend and AI-related processes.
- **Processor:** A processor like Intel® Core™ i5 or AMD Ryzen 5 (or higher) is preferred so that both browser and backend operations can run without lag.
- **Hard Disk:** Around 20 GB of free space is needed to store project files and related data. Both HDD and SSD can be used, although SSD is faster.
- **Graphics Processing Unit:** A GPU is not compulsory, but if available, it can speed up processing for transformer models. Otherwise, the system can still run using CPU.
- **Internet Connectivity:** A stable internet connection is necessary to use Overleaf and also to fetch research papers using APIs like CrossRef.

These requirements are generally enough for the system to run properly and handle the required text processing tasks without much difficulty.

3.1.2 Software Requirements

The software environment includes the operating system, programming tools, frameworks, and libraries needed for developing and running the proposed system. Each component has a specific role to play in facilitating the functionality of the WriteSense AI platform.

- **Operating System:** Windows 10/11, Ubuntu 20.04 LTS, and later versions.
- **Web Browser:** Google Chrome, which supports Chrome extensions and allows for the integration of the Overleaf editor.
- **LaTeX Authoring Platform:** Overleaf, a web-based LaTeX editor for writing academic content.
- **Programming Language:** Python 3.9 and later versions for backend implementation and AI model integration.
- **Backend Framework:** FastAPI, used for developing modular backend services for handling user requests.
- **Backend Server:** Uvicorn ASGI server, used for running FastAPI-based backend services.
- **Grammar Checking Tool:** LanguageTool, used for rule-based grammar checking.
- **AI Model Runtime:** Ollama, used for running large language models locally.
- **Machine Learning Framework:** Hugging Face Transformers, used for developing summarization and translation models.

The software requirements for developing the AI-based writing assistance tool are listed above.

3.1.3 *Software Requirements for Development*

The software requirements for development include various programming tools, frameworks, and libraries used during the implementation phase of the project. These development tools will help the developer in developing the different modules of the WriteSense AI-based system.

- **Programming Language:** Python 3.9 is used for backend service development and AI model integration.
- **Frontend Technologies:** JavaScript, HTML, and CSS are used for developing the Chrome extension interface.
- **Browser Extension Standard:** Chrome Extension Manifest V3 is used for developing Chrome extensions.
- **Development Environment:** Visual Studio Code is used as the development environment.
- **Backend Server:** Uvicorn ASGI server is used to run the FastAPI application.
- **Machine Learning Libraries:** The Transformers and PyTorch libraries are used for natural language processing.
- **AI Models Used:**
 - LLaMA Model via Ollama for grammar enhancement and write improvement.
 - NLLB Multilingual Translation Model (*facebook/nllb-200-distilled-600M*) for language translation.
- **API Communication:** RESTful API using the Fetch API in JavaScript and the Requests library in Python.
- **Research Paper Retrieval:** CrossRef API for accessing publication metadata.

This software development environment facilitates the integration of the browser-based interfaces with the backend processing modules.

3.2 Software Specification

The software specification section explains the programming languages and tools that were used to design and develop the proposed WriteSense AI system. The system mainly combines front-end technologies with backend services to make everything work together smoothly. It also includes rule-based grammar correction methods, transformer-based natural language processing models, and tools for recommending research papers.

All these software components work together to provide useful writing support to the user. At the same time, care has been taken to ensure that the original LaTeX structure is not affected. This helps in maintaining both the correctness of the syntax and the overall academic quality of the document.

3.2.1 Python 3.9

The proposed system uses Python 3.9 for developing the backend part of the application. Python was mainly selected because it is easy to use and also provides good support for web development, machine learning, and natural language processing.

In this system, Python is used to build different AI-based features like grammar correction, improving clarity, summarizing text, translating content, and suggesting related research papers. These features are important for providing writing support to the user. Python is also used to manage the communication between the Chrome extension and the backend through RESTful APIs. This helps in sending the selected text to the server and getting the processed result back in a simple way. Overall, Python is a suitable choice for this project because of its flexibility and the availability of many useful libraries, which makes development easier. It also enables rapid development and easy integration with external APIs and AI models. Furthermore, Python's strong community support and extensive

documentation make it easier to debug, maintain, and extend the system in future enhancements.

3.2.2 *FastAPI*

The backend web framework used in the implementation of the modular AI services is FastAPI. Each significant feature of the system, such as grammar correction, clarity analysis, research paper detection, summarization, and translation, has been implemented as a separate FastAPI service. The use of the asynchronous request handling feature of the framework ensures the processing of concurrent requests efficiently. This ensures the system can handle multiple user operations without compromising performance.

3.2.3 *LanguageTool*

The system uses the LanguageTool tool as the backend engine for the implementation of the rule-based grammar correction feature. The tool has the capability to detect grammatical mistakes such as spelling mistakes, punctuation mistakes, subject-verb agreement mistakes, and mistakes in the use of articles. The use of the LanguageTool tool ensures reliable rule-based grammar correction in the system. The grammatically correct text maintains the original meaning and academic tone of the text.

3.2.4 *Large Language Models and Transformer Models*

The system uses several transformer-based AI models to implement the advanced writing assistance feature. The system uses the LLaMA language model running on the Ollama runtime for the implementation of the structural grammar correction feature and the academic sentence correction feature. The system uses the NLLB model (*facebook/nllb-200-distilled-600M*) for the implementation of the translation feature. The model supports the translation of text into more than forty languages. The fact that the model makes use of the attention mechanisms to analyze the relationship

between the different words in the entire sentence makes the system able to comprehend the context and provide accurate linguistic feedback.

3.2.5 *CrossRef API*

For the research paper recommendation service, the system makes use of the CrossRef API. This API gives access to millions of publications. The system extracts research-related keywords from the chosen text and uses the CrossRef API to retrieve relevant research papers. The retrieved results include the title, authors, publication year, and DOI links to the research papers. These results enable the user to easily identify relevant research papers related to the research field.

3.2.6 *JavaScript and Chrome Extension APIs*

JavaScript has been used to develop the Chrome extension to integrate the AI-based writing system with the Overleaf editor. Chrome Extension APIs enable the content script to access and interact with the Document Object Model (DOM) of the webpage and retrieve the selected text. The Fetch API has been used to send requests to the backend service. Additionally, the extension interface has been used to provide task-specific buttons such as grammar correction, clarity, research paper, summarization, and translation.

3.2.7 *JSON-Based Structured Communication*

For the communication between the Chrome extension and the backend service, the system makes use of structured communication using JSON. This ensures the reliable parsing of the results and prevents the unintended modification of the LaTeX command structures. The software environment for the proposed WriteSense AI system consists of browser-based technologies, Python-based backend microservices, rule-based grammar correction tools, transformer architecture-based natural language processing tools, and metadata retrieval tools for scholarly information. Grammar correction is achieved through LanguageTool

and AI-based models, while summarization and translation are achieved through transformer architecture.

Integration with Overleaf editor via a Chrome extension will enable a smooth workflow and user-centric writing assistance. This combination will yield a robust, modular, and scalable framework for improving writing standards while maintaining the document structure in LaTeX format.

In summary, this chapter discussed the software environment and technical requirements necessary for the development and operation of the WriteSense AI system. The next chapter, System Analysis, will outline the various modules and workflow of the proposed system, provide a detailed examination of its requirements, and explain how WriteSense AI addresses the challenges of academic writing in LaTeX environments.

CHAPTER 4

SYSTEM ANALYSIS

System analysis implies a thorough investigation of the proposed WriteSense AI system aimed at the evaluation of the system's goals, functions, and processes. The main aim of this analysis lies in assessing the efficiency of the use of the system in the context of helping users with academic writing in a LaTeX environment by ensuring an integrated linguistic assistance without breaking the process of writing. System analysis aims to explore such aspects of the system as a user interaction with a Chrome extension, the processing of a selected text, and generation of feedback information.

Additionally, the functional division between grammar checking according to certain rules and sentence improvement using artificial intelligence, the use of clarity analysis for the evaluation of writing, summarization capabilities, multilingual translation, and research paper recommendations as the components important for academic writing are going to be analyzed. These factors will be considered from the perspective of their contribution to the preservation of semantic meaning, LaTeX syntax, and user control over text alterations.

4.1 Requirements Analysis

Requirements analysis is a formal method aimed at discovering and defining the functional and non-functional needs of the suggested system. The set of requirements states the anticipated performance of the system, services it must offer, and limitations that will govern its operation in the context of user interaction. Knowledge about these requirements guarantees

that the system functions as expected and fulfills its purpose efficiently.

In this project proposal, requirements are based on the need for text selection and processing by an AI agent within a LaTeX document editor. The system should be able to perform grammar correction, academic clarity verification, summarize the text, translate into multiple languages, and recognize related literature. Additionally, the system must preserve semantic meaning, maintain LaTeX syntax, and give the user complete control over all changes made to the text.

4.2 Existing System

Presently, authoring tools like Overleaf that make use of LaTeX have become quite popular in academic settings as far as the creation of technical and scientific papers is concerned. Overleaf has robust facilities for authoring in LaTeX, which include document structuring, mathematical equations, citations, versions control, and collaborative work on the document.

Nevertheless, even though Overleaf offers great facilities for document structuring and collaborative work, it does not provide any language facilities for enhancing the quality of writing.

4.3 Limitations of Existing System

Despite the widespread adoption of LaTeX-based authoring platforms like Overleaf in academic and technical writing, several limitations hinder their effectiveness as comprehensive writing tools. Understanding these shortcomings is crucial for motivating the need for an integrated AI-based solution. The primary limitations of existing systems are as follows:

- Overleaf does not support any native functionality for grammar checking, readability, or assistance for academic writing.
- The authors have to depend upon any external tool for grammar checking or AI writers to make their content linguistically better.

- External tools force users to manually copy and paste the LaTeX text content in and out of the editor.
- There are chances that the syntax can get corrupted while working with the LaTeX text content in external tools.
- Text formats, macros, and LaTeX commands can be lost while working in external environments.
- None of the current LaTeX writing tools supports any summarization or translation of academic texts.
- Literature finding activities are often done externally, which disrupts the writing process.
- Most of the external AI writer tools carry out automatic text rewriting without offering any transparency to the users.
- Privacy issues may come due to sending sensitive research data to cloud-based AI writers.

4.4 Proposed System

WriteSense AI is an advanced browser extension designed specifically for integration with the Overleaf editor in Google Chrome. It addresses the key challenges faced by academic writers working in LaTeX environments by providing intelligent, context-aware writing support directly within the editor. The tool offers a comprehensive suite of features, including grammar correction, clarity assessment, summarization, multilingual translation, and related literature recommendations, all accessible through an intuitive user interface.

The system is built on a modular architecture, with each major functionality implemented as an independent FastAPI service. This design ensures scalability, maintainability, and ease of future expansion. When a user selects text in the Overleaf editor, the Chrome extension communicates

with the backend services to analyze and process the content. Advanced AI algorithms and linguistic techniques are applied to deliver precise and contextually appropriate feedback, while preserving important LaTeX structures such as commands, formulas, and citations.

A key principle of WriteSense AI is user control and transparency. All suggestions and modifications are presented to the user for review before any changes are applied, ensuring that the academic integrity and intent of the document are maintained. This approach empowers users to iteratively improve their writing quality without sacrificing document structure or workflow efficiency.

By seamlessly integrating powerful AI-driven assistance within the familiar Overleaf environment, WriteSense AI streamlines the academic writing process and enables users to produce clearer, more polished, and professional documents with greater confidence and less effort.

4.5 Advantages of Proposed System

After examining the limitations of existing LaTeX writing tools and identifying the requirements for effective academic writing assistance, it is essential to propose a solution that addresses these challenges. The following section introduces the WriteSense AI system, detailing its key features, advantages, and how it overcomes the shortcomings of current approaches.

- Integrating AI writing suggestions within the Overleaf LaTeX editor itself.
- No reliance on external writing tools which helps keep the writing flow intact.
- Performing AI-assisted grammar checking and rule-based grammar checking through LanguageTool to provide deterministic corrections.
- Linguistic and readability analysis of the text using parameters like lexical density, passive voice percentage, etc.

- Summarizing the selected text so that the writer can make improvements.
- Translating academic texts into various languages without breaking LaTeX commands.
- Retrieval of relevant research papers through keyword matching and academic metadata mining.
- Preventing corruption of LaTeX commands during processing with LaTeX command-sensitive preprocessing.
- Giving users full control over their text with the need for their consent before making any change.

4.6 Functional Requirements

Functional requirements explain what the system is expected to do and what kind of services it should provide to the user.

4.6.1 User Interface

The system will provide a simple and easy-to-use interface in the form of a floating toolbar inside the Overleaf editor. This toolbar will include options like Grammar Correction, Clarity Analysis, Research Papers, Summarization, and Translation so that users can easily access different features.

4.6.2 User Input

The user will have the flexibility to select any specific sentence, paragraph, or section of interest within the LaTeX editor. Upon making a selection, the system will focus exclusively on the highlighted portion, ensuring that processing is both targeted and relevant. This selective approach not only enhances efficiency but also allows users to refine their writing incrementally without affecting the rest of the document.

4.6.3 Grammar Correction

The system will identify and correct grammar mistakes such as tense errors, subject–verb agreement, spelling, and punctuation. This will be done using tools like LanguageTool along with AI models like LLaMA.

4.6.4 Clarity Analysis

The system will check how clear and readable the text is by using measures like Flesch Reading Ease, lexical density, clause density, and passive voice ratio. This helps users understand and improve the quality of their academic writing.

4.6.5 Summarization of Selected Text

The system will generate a concise summary of the selected text, capturing the main ideas and essential information without introducing any new content. This feature enables users to quickly review and understand the key points of their writing, streamlining the process of content refinement.

4.6.6 Multilingual Translation

The system will translate the selected text into the user’s desired language, enabling seamless multilingual communication within academic documents. During the translation process, the system ensures that all LaTeX commands, citations, and mathematical equations remain intact and unaltered. This careful preservation of formatting allows users to work confidently across multiple languages without compromising the structure or integrity of their technical content.

4.6.7 Related Research Paper Identification

The system will intelligently extract key concepts and keywords from the selected text to facilitate efficient literature discovery. Using these keywords,

it will search for related research papers through the CrossRef API, ensuring that the recommendations are contextually relevant. The results will include essential publication details such as the title, authors, year of publication, and DOI links, making it easier for users to explore and reference pertinent academic work.

4.6.8 Apply and Control Mechanism

The system will not make any automatic changes to the document. Users must approve all suggested corrections or improvements, which helps maintain control and transparency.

4.7 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the WriteSense AI academic writing system. Unlike functional requirements, which specify what the system does, non-functional requirements describe how effectively the system performs its tasks. These requirements ensure that the system delivers accurate feedback, maintains efficiency, and provides a smooth user experience. For the proposed AI-based writing assistant, aspects such as performance, usability, reliability, and scalability are essential to support grammar correction, readability analysis, AI-based evaluation, and multilingual translation features.

4.7.1 Performance

Performance refers to the system's ability to process textual data efficiently and provide rapid feedback. The WriteSense AI system must analyze user input for grammar errors, readability metrics, tone consistency, and translation within a short response time. Since the system uses natural language processing and machine learning models, it should optimize CPU usage and memory consumption to ensure smooth execution. The system

should be capable of handling multiple user requests simultaneously while maintaining consistent response speed and accuracy.

4.7.2 Usability

Usability defines how easily users can interact with the system and understand its outputs. The system should provide a simple and intuitive interface where users can input or upload text and receive clear feedback. Results such as grammar suggestions, readability scores, AI evaluation insights, and translated content should be presented in a structured and user-friendly manner. The interface design should minimize complexity, allowing students, researchers, and writers to effectively use the system without requiring technical expertise.

4.7.3 Compatibility

Compatibility ensures that the system operates seamlessly across different platforms and environments. The WriteSense AI system should be accessible through modern web browsers and support multiple operating systems such as Windows, Linux, and macOS. It should integrate with backend technologies such as Python and machine learning libraries, and support various input formats including plain text and document files. This flexibility enables users to access the system from different devices and environments without compatibility issues.

4.7.4 Scalability

Scalability refers to the system's ability to handle increasing workloads and extend its functionalities over time. The WriteSense AI system should be capable of processing larger volumes of text and supporting a growing number of users without performance degradation. It should also allow future enhancements such as advanced AI models, additional language support, plagiarism detection, and real-time collaborative editing. This ensures that the system remains adaptable and efficient.

4.7.5 Reliability

Reliability ensures that the system consistently performs its intended functions without failure. The WriteSense AI system should provide accurate grammar corrections, consistent readability analysis, and dependable translation outputs. It should handle errors gracefully and ensure minimal downtime during operation. Regular updates and model improvements should be incorporated to maintain accuracy and system stability over time.

4.7.6 Security

Security refers to the protection of user data and system resources from unauthorized access. The system should ensure that all user-submitted text is handled securely and not exposed to third parties. Appropriate measures such as data encryption, secure communication protocols, and access control mechanisms should be implemented. This is especially important as users may input sensitive academic or research content into the system.

4.8 Feasibility Study

The feasibility study evaluates whether the system can be developed and implemented effectively with the available technology, resources, and within the given time frame. It examines the project from technical, operational, and economic perspectives to ensure its overall viability.

4.8.1 Technical Feasibility

The system is technically feasible because it leverages established and widely adopted technologies such as Chrome extensions, FastAPI, grammar checking tools, and advanced NLP models. These technologies provide a stable foundation for development and integration, reducing the risk of technical obstacles. Additionally, the modular design of the system simplifies the implementation, maintenance, and future expansion.

4.8.2 Operational Feasibility

The system will work directly in a web browser and integrate smoothly with Overleaf. Since it uses simple features like text selection and toolbar buttons, it will be easy for users to understand and use without much training.

4.8.3 Economic Feasibility

The system does not require expensive hardware and mainly uses free or open-source tools like FastAPI, LanguageTool, and AI models. This makes it cost-effective and suitable for academic use.

This chapter explained the functional and operational aspects of the WriteSense AI system. It also discussed the problems in existing LaTeX writing tools, such as lack of built-in writing support, dependency on external tools, and possible errors in LaTeX formatting. The proposed system solves these issues by providing AI-based writing help directly in Overleaf. It improves writing quality while keeping the document structure safe and giving full control to the user. The feasibility study shows that the system is practical and can be successfully implemented.

The next chapter will discuss the system design of WriteSense AI, including its architecture and how the Chrome extension communicates with backend services for different features like grammar correction, clarity analysis, summarization, translation, and research paper retrieval.

CHAPTER 5

SYSTEM DESIGN

This chapter provides a comprehensive overview of the system design for the proposed LaTeX WriteSense AI platform. It covers both the overall architecture and the detailed data flow mechanisms that enable seamless integration and processing within the system. The chapter explains how each component interacts, ensuring effective delivery of writing assistance features while maintaining the integrity of LaTeX documents.

5.1 System Architecture

Figure 5.1 presents the architecture for the proposed LaTeX WriteSense AI system. The LaTeX WriteSense AI system architecture presents how the user can select LaTeX text within the Overleaf editor and how the selected LaTeX text will be processed within different language processing modules for providing grammar correction, clarity improvement, summarization, multilingual translation, and identification of related research papers. The LaTeX WriteSense AI system will maintain LaTeX syntax while providing all these facilities to the user.

The LaTeX WriteSense AI system will be developed based on client-server architecture. The Chrome extension integrated with Overleaf will act as a client for the LaTeX WriteSense AI system. The backend services developed using FastAPI will act as a server for the LaTeX WriteSense AI system. When a user selects a portion of LaTeX text within the Overleaf editor, the Chrome extension integrated into Overleaf will forward the selected LaTeX text to the backend server as a JSON request.

The LaTeX WriteSense AI system will have a backend processing layer

containing different modules for processing LaTeX text for providing different language processing facilities. The grammar processing module uses rule-based grammar checking with the help of the LanguageTool tool and grammatical improvement with the help of the LLaMA model. The clarity analysis module uses large language models for evaluating the text's readability and structural quality. The summarization module uses the LLaMA model for summarizing the chosen text. The translation module uses the NLLB model for translating the text into different languages. The research paper identification module identifies the important keywords in the text and fetches relevant research papers from the CrossRef API.

Once the text has undergone all the processes, the backend server sends the output back to the Chrome extension in JSON format. The Chrome extension then displays the output and explanation to the user. The user can then review the output and decide whether they want to accept it or reject it before it is applied to the LaTeX document. This way, the user has full control over the document.

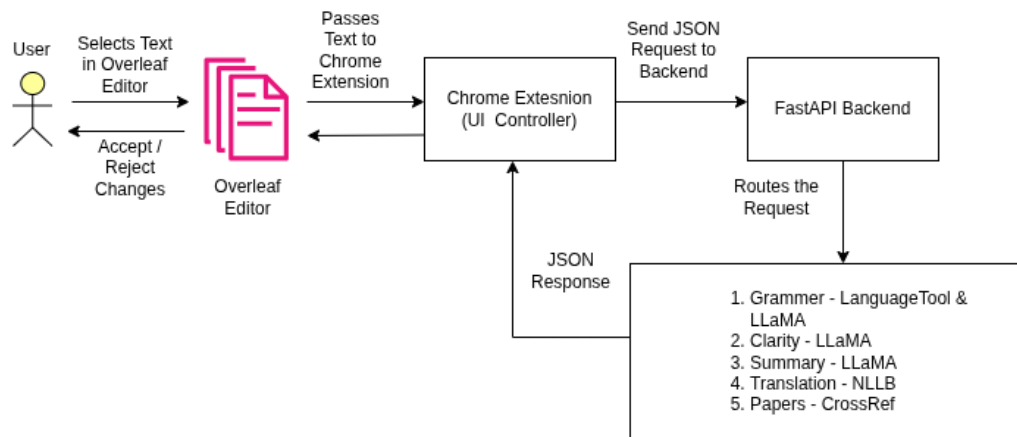


Fig. 5.1: Architecture Diagram

5.2 Data Flow Diagram

The Data Flow Diagram shows the movement of data in the LaTeX WriteSense AI tool from the input to the output. It shows that the tool processes the text chosen by the user from the Overleaf editor. Furthermore, the selected input is sent to the backend processing module,

where it undergoes analysis and refinement using AI-based models. The processed output is then returned to the editor, ensuring that the original structure and LaTeX formatting are preserved. This flow illustrates how data is handled efficiently across different components, enabling seamless interaction between the user interface and the underlying AI system.

5.2.1 DFD Level 0

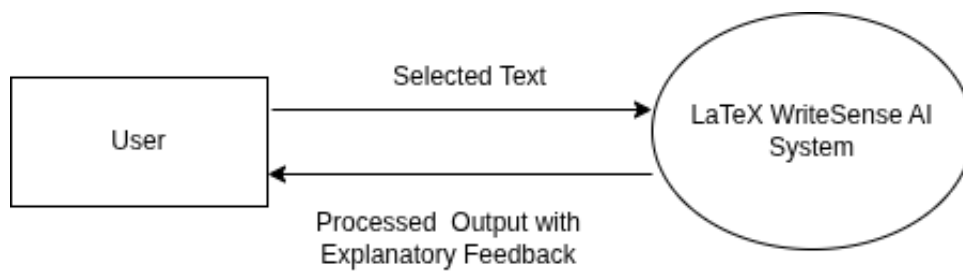


Fig. 5.2: Data Flow Diagram Level 0

Figure 5.2 shows the Data Flow Diagram (DFD) Level 0 of the proposed LaTeX WriteSense AI system. This level indicates the high-level view of the system, which shows the interaction between the user and the processing framework without revealing the inner workings of the modules.

At this level, the user is prompted to provide inputs by selecting the text inside the Overleaf editor. This selected text is then passed to the LaTeX WriteSense AI system. This system is responsible for performing specific tasks, including grammar correction, clarity, summarization, translation, and research paper detection.

After this, the system will provide processed information and feedback to the user. This information is then passed back to the user through the Chrome extension interface. Finally, the LaTeX document is modified based on the changes made by the system. This is done only with the approval and consent of the user. This is important to ensure transparency and academic integrity. Additionally, the system ensures that all modifications preserve the original meaning and LaTeX structure of the document. This controlled workflow enhances user trust while maintaining consistency and reliability in the editing process.

5.2.2 DFD Level 1

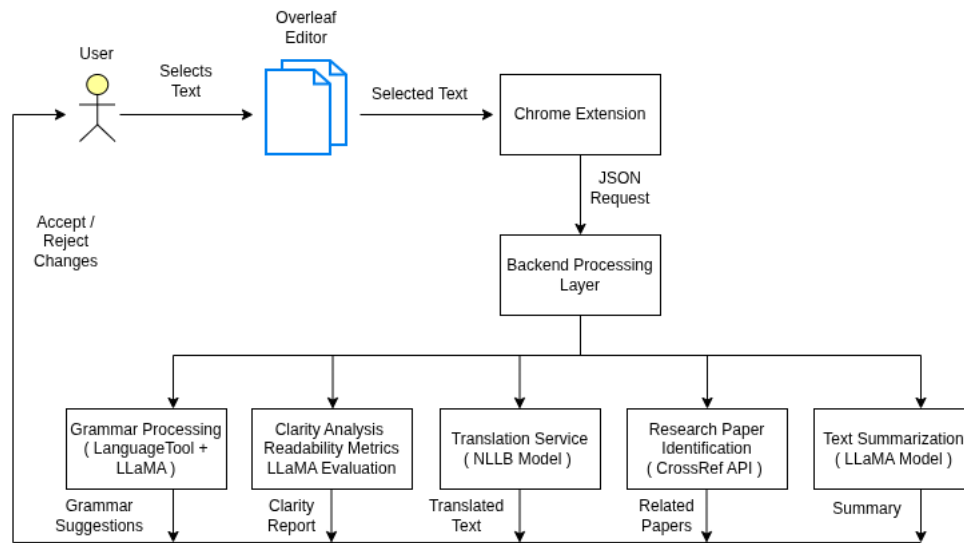


Fig. 5.3: Data Flow Diagram Level 1

Figure 5.3 represents the Data Flow Diagram (DFD) Level 1 of the proposed LaTeX WriteSense AI system. This level gives an idea about the internal data processing by showing the decomposition of the system into various modules.

The process begins with the selection of the text by the user inside the Overleaf editor. This selected text is then captured by the Chrome extension, which is the interface component of this system. This Chrome extension then sends the selected text to the system by sending a JSON request to the FastAPI server. The backend processing layer is in charge of directing the request to the corresponding processing module depending on the selected operation by the user. The grammar processing module carries out grammatical analysis of the text using the LanguageTool tool. The clarity analysis module uses AI-based techniques to evaluate the text. The summarization module generates the summary of the selected text. The translation service uses the NLLB model to translate the selected text into the target language. The research paper identification module uses the CrossRef API to fetch the required academic research papers. After processing the request by the processing modules, the results obtained from

the processing modules are sent back to the Chrome extension in the form of JSON responses. The Chrome extension then displays the results obtained from the processing modules along with the required information. The user then has the complete freedom to apply the suggested changes to the LaTeX document or not. The user has complete freedom in this regard because the changes are applied in a transparent manner.

This chapter has outlined the design of the proposed LaTeX WriteSense AI system, detailing its overall architecture, data flow, and the functional modules that comprise the platform. It has also explained how the system interacts with both the Chrome extension interface and the backend services developed using FastAPI, ensuring seamless integration between frontend and backend components. Special attention has been given to preserving the integrity of LaTeX documents and maintaining user control throughout the editing process. The following chapter will focus on System Implementation, describing how the proposed design is brought to life through development, module integration, and the practical realization of the WriteSense AI platform.

CHAPTER 6

SYSTEM IMPLEMENTATION

The system implementation phase marks the transformation of the conceptual design of the WriteSense AI system into a fully functional academic writing assistance tool. In this stage, the modules defined during the design phase are developed, integrated, and thoroughly tested to ensure that all intended functionalities operate effectively. The implementation brings together rule-based grammar correction, AI-based grammar correction, clarity evaluation, text summarization, multilingual translation, and research paper identification within a unified Chrome extension platform.

The primary objective of this phase is to create a practical system that enables researchers and students to enhance the quality of their academic writing directly within a LaTeX editing environment. The system is realized as a Chrome extension that interacts seamlessly with backend services responsible for natural language processing and linguistic analysis. To promote scalability and modularity, the WriteSense AI system employs a microservices-based architecture. Each feature is developed as an independent processing module, communicating with others through REST APIs. This design ensures that individual components can be updated or maintained without disrupting the overall system.

The implementation phase integrates several key subsystems that deliver comprehensive academic writing assistance.

- Grammar Correction Engine
- Academic Clarity Analysis Engine

- Text Summarization Module
- Multilingual Translation Module
- Research Paper Identification Engine

Each subsystem interacts with backend services developed using the FastAPI framework and incorporates external APIs and AI models for advanced text processing.

6.1 Development Environment Setup

The development of the WriteSense AI system involves a combination of frontend technologies, backend frameworks, and natural language processing models. The following tools and technologies were used during the implementation phase.

- Frontend: JavaScript (Chrome Extension using Manifest V3)
- Backend: Python 3.10
- Framework: FastAPI
- Backend Server: Uvicorn ASGI Server
- AI Model Runtime: Ollama (LLaMA3)
- Grammar Checking Tool: LanguageTool
- Translation Model: NLLB (facebook/nllb-200-distilled-600M)
- Research API: CrossRef
- Development Environment: Visual Studio Code

The backend services are deployed locally using Uvicorn servers running on separate ports for grammar processing, clarity analysis, summarization, translation, and research functionalities. Each service operates independently while communicating with the Chrome extension

via REST API calls. To enable seamless communication between the extension and backend services, Cross-Origin Resource Sharing (CORS) middleware is configured within the FastAPI framework. This setup allows secure data exchange between frontend and backend components.

6.2 Chrome Extension Implementation

The frontend of WriteSense AI is implemented as a Chrome extension that integrates directly with the Overleaf LaTeX editor. The extension injects content scripts into the Overleaf interface, enabling interaction with the editor and capturing user-selected text. The extension performs several essential operations to support real-time academic writing assistance.

- Detects user-selected text within the LaTeX editor.
- Sends asynchronous API requests to backend services.
- Displays analysis results and improvement suggestions in a dock-style interface.
- Allows users to review, accept, or reject suggested changes before applying them.

The user interface is dynamically generated using JavaScript template strings and styled with CSS components. This ensures that the writing workflow remains uninterrupted while still providing easy access to assistance features.

6.3 Grammar Correction Engine

The grammar correction engine is designed to detect and rectify grammatical errors present in user-selected text within the LaTeX editor. It achieves this by combining robust rule-based approaches with advanced AI-driven enhancements, ensuring that corrections are both precise and sensitive to contextual nuances. This integrated methodology not only

increases grammatical accuracy but also significantly improves the overall readability and academic tone of the writing.

6.3.1 Rule-Based Grammar Correction

In the initial stage of grammar analysis, the selected text is sent to the grammar check endpoint of the backend service. Several preprocessing steps are performed before analysis.

- Protection of LaTeX commands and citation structures
- Detection of grammar and spelling errors
- Integration with the LanguageTool API
- Generation of correction suggestions with positional offsets

Each detected issue includes details such as the incorrect word, its position in the sentence, the suggested correction, and the type of grammatical error.

6.3.2 AI-Based Structural Grammar Improvement

In addition to rule-based correction, the system applies AI-based techniques to further refine grammatical accuracy in the selected text. The text is sent to the grammar AI-improve endpoint, which performs the following steps:

- Protecting LaTeX commands and references to prevent unintended modifications
- Generating a grammar correction prompt tailored for academic writing
- Processing the prompt using the LLaMA3 model through the Ollama runtime to identify and correct grammatical errors
- Cleaning and formatting the AI-generated output to ensure it aligns with LaTeX conventions

The AI-generated output corrects grammar mistakes and inconsistencies while maintaining the original meaning and academic tone of the text.

6.4 Academic Clarity Analysis Engine

The clarity analysis module evaluates the readability and academic quality of selected text using both statistical methods and AI-based techniques.

6.4.1 Readability Metrics

The system computes standard readability metrics, including the following:

- Flesch Reading Ease (FRE)
- Flesch–Kincaid Grade Level (FKG)

These metrics are derived from sentence count, word count, and syllable count, providing a quantitative assessment of text readability.

Flesch Reading Ease (FRE): This metric is computed by segmenting the input text into sentences, determining the total number of words, and estimating the number of syllables in each word. These components are analyzed to evaluate the ease with which the text can be understood.

Flesch–Kincaid Grade Level (FKG): This metric is obtained using the same linguistic features, namely sentence count, word count, and syllable count. The computed values are used to estimate the educational grade level required for comprehension of the text.

6.4.2 Structural Academic Metrics

In addition to readability, the system evaluates several structural indicators commonly used in academic writing:

- Passive Voice Percentage
- Lexical Density

- Clause Density
- Nominalization Ratio

These measures provide insights into the structural complexity and informational richness of the text.

Passive Voice Percentage: This metric is determined by identifying sentences expressed in passive voice using grammatical analysis techniques and computing their proportion relative to the total number of sentences.

Lexical Density: This measure is computed by identifying content-bearing words, including nouns, verbs, adjectives, and adverbs, and comparing their frequency with the total word count to assess the informational density of the text.

Clause Density: This metric is obtained by decomposing sentences into constituent clauses through syntactic parsing and comparing the total number of clauses with the number of sentences to evaluate structural complexity.

Nominalization Ratio: This measure is calculated by detecting nominalized forms derived from verbs or adjectives (e.g., words with suffixes such as -tion, -ment, and -ness) and comparing their occurrence with the total word count, reflecting the degree of formal academic expression.

6.5 Text Summarization Module

The summarization module generates concise summaries of academic text using transformer-based models. The selected text is processed by the backend service using the LLaMA model through the Ollama runtime. The resulting summary captures the key ideas and essential information while maintaining the original academic context. Additionally, the module ensures coherence and logical flow in the summarized output, making it easier to understand complex content. This functionality helps users quickly grasp the main contributions of lengthy academic material without compromising clarity or meaning. Furthermore, the module is designed

to preserve key terminologies and domain-specific expressions, ensuring that the summarized output remains technically accurate. It also reduces redundancy by eliminating repetitive information.

6.6 Multilingual Translation Module

The translation module enables multilingual translation of academic text using the NLLB model. The selected text and target language are sent to the backend service, where the translation is generated.

The process preserves the semantic meaning of the text while ensuring that LaTeX commands, mathematical expressions, and citations remain unchanged.

6.7 Research Paper Identification Module

The research paper identification module helps users discover relevant academic literature related to the selected text. The system analyzes the text to extract key concepts and keywords using AI-based techniques. These keywords are then used to query the CrossRef API, which provides access to scholarly publications. The retrieved results include titles, authors, publication years, and DOI links. These are formatted in IEEE citation style and displayed within the extension interface.

6.8 System Integration

All modules communicate through asynchronous REST API requests between the Chrome extension and FastAPI-based backend services. The extension sends user-selected text, and the backend returns structured JSON responses containing analysis results. This modular design ensures scalability, maintainability, and efficient processing while preserving LaTeX document integrity.

This chapter presented the implementation of the WriteSense AI system in detail. The system integrates a Chrome extension frontend with

multiple FastAPI-based backend services to provide real-time academic writing assistance. The grammar correction module combines rule-based detection using LanguageTool with AI-based improvements using the LLaMA3 model, while the clarity analysis module evaluates writing quality using readability and structural metrics. Additionally, the system supports text summarization, multilingual translation using the NLLB model, and research paper identification through the CrossRef API. Each subsystem operates independently within a modular architecture. The combination of rule-based techniques, statistical analysis, and AI-driven processing provides a comprehensive solution for academic writing assistance. The next chapter presents the experimental results and performance evaluation of the system.

CHAPTER 7

RESULTS AND DISCUSSION

This chapter presents the experimental results and functional evaluation of the proposed WriteSense AI system. The system was tested in a real academic writing environment using the Overleaf LaTeX editor. The evaluation focuses on the effectiveness of grammar correction, clarity analysis, text summarization, multilingual translation, and research paper identification modules.

The experimental analysis was performed by applying the system to various academic text samples including literature review paragraphs, technical descriptions, and research abstracts. These experiments were conducted to observe how effectively the system detects linguistic issues, evaluates readability, generates summaries, translates content, and identifies related research papers.

The experimental results demonstrate that the proposed system successfully integrates multiple AI-based writing assistance tools within a single Chrome extension interface. By embedding these functionalities directly within the Overleaf editor, the system enables seamless academic writing support without disrupting the authoring workflow. The modular architecture ensures that each component performs its task independently while maintaining overall system efficiency.

7.1 *System Interface Overview*

Figure 7.1 illustrates the main interface of the WriteSense AI extension integrated with the Overleaf editor. The system appears as a set of floating action buttons labeled *Grammar*, *Clarity*, *Papers*, *Summary*, and *Translate*.

These buttons allow users to activate different writing assistance modules directly within the editor.

The interface is designed to operate without interrupting the writing workflow. After selecting a portion of text, users can activate any module to analyze or process the selected content. The system then displays the results in a dock-style panel on the right side of the screen.

The floating interface design ensures that the user can easily access system features without modifying the structure of the LaTeX document. The dock panel provides organized sections for displaying analysis results, allowing users to review suggestions before applying them to the document.

This interface confirms the successful integration of the Chrome extension with the Overleaf editor. The floating button design ensures accessibility, ease of use, and modular interaction while preserving the layout of the writing environment. The interface design also enhances usability by reducing the need for external tools during the writing process. Furthermore, it streamlines the workflow by enabling seamless in-editor assistance without interrupting the user’s writing experience.

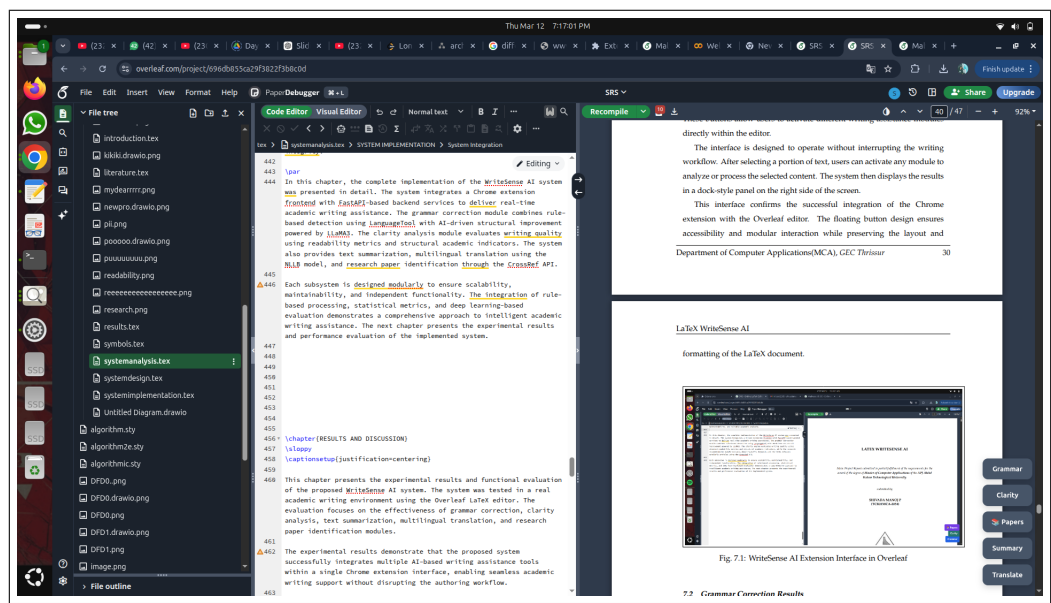


Fig. 7.1: WriteSense AI Extension Interface in Overleaf

7.2 Grammar Correction Results

Figure 7.2 illustrates the results produced by the Grammar Correction Module, which incorporates both rule-based and AI-based grammar correction methods to maximize error detection. These two approaches function in a complementary manner, each capable of identifying certain errors the other might miss.

The rule-based grammar correction engine, powered by LanguageTool, is adept at detecting explicit grammatical mistakes such as incorrect verb forms, punctuation errors, spelling mistakes, and subject-verb agreement issues. It systematically reviews the text and provides detailed explanations and targeted suggestions for each detected problem.

The AI-based module, utilizing the LLaMA language model, is designed to detect grammatical errors as well, including those that may be context-dependent or less obvious to traditional rule-based systems. While it does not enhance sentence structure or style, the AI-based approach excels at identifying subtle or complex grammar issues that might escape rule-based detection. There are many instances where the AI-based module catches errors missed by the rule-based system, and vice versa, highlighting the strengths of combining both methods.

Both modules use an offset-based correction mapping technique, ensuring that only the intended segments of text are altered, with all LaTeX commands and adjacent content preserved.

By comparing the outputs from both rule-based and AI-based modules, users receive a broader and more comprehensive set of grammar suggestions. This dual-method approach results in writing that is more accurate and free from overlooked grammatical mistakes. The combined methods enhance user confidence and control, helping writers produce clearer, error-free, and academically sound documents while maintaining the integrity of their LaTeX formatting. This comprehensive approach ultimately streamlines the revision process and improves the overall quality of academic writing.

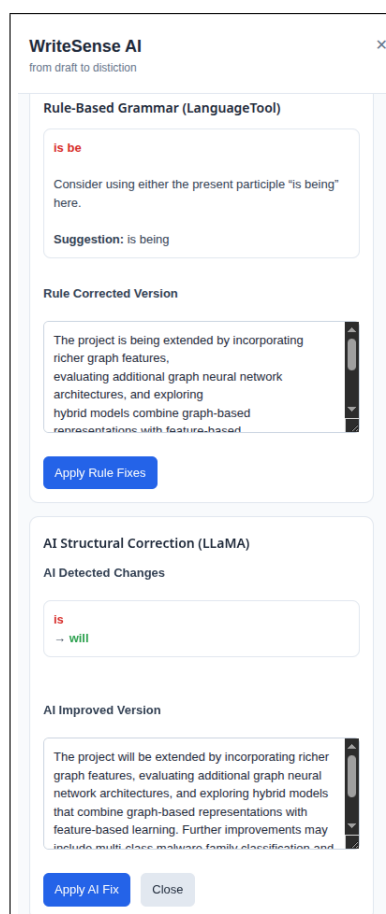


Fig. 7.2: Grammar Correction Module Output

7.3 Clarity Evaluation Results

Figure 7.3 presents the output of the Clarity Analysis Module. The system evaluates the readability and structural quality of the selected academic text using both statistical readability metrics and AI-based evaluation.

The module calculates the Flesch Reading Ease (FRE) and Flesch–Kincaid Grade Level (FKG) using standard readability formulas. Experimental results indicate that the analyzed text corresponds to advanced academic writing levels typically found in postgraduate research papers.

The system also computes structural academic indicators including passive voice percentage, lexical density, clause density, and nominalization ratio. These metrics help determine the complexity and abstraction level of

the writing. High lexical density values indicate information-rich academic writing, while moderate nominalization ratios reflect appropriate levels of abstraction commonly found in scholarly texts. Passive voice analysis also helps authors understand whether their writing style aligns with academic conventions.

Additionally, the integration of quantitative readability metrics with qualitative AI-driven evaluation highlights the robustness of the clarity module in delivering comprehensive insights into academic writing quality. By combining objective measures with contextual language analysis, the system is able to identify both surface-level and deeper structural issues within the text. This dual approach ultimately enables more precise, actionable feedback that supports the development of clear, coherent, and academically rigorous writing. Furthermore, this integrated methodology enhances the reliability of assessments, ensuring that writers receive balanced and well-informed guidance for continuous improvement.

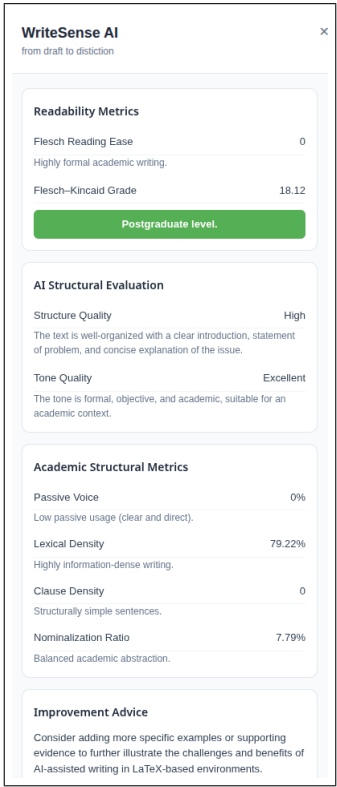


Fig. 7.3: Clarity Analysis Module Output

7.4 Text Summarization Results

Figure 7.4 illustrates the output of the Text Summarization Module. The module generates a concise summary of the selected academic text while preserving its core meaning and key information.

The summarization process is performed using a transformer-based language model operating through the backend service. The system analyzes the semantic structure of the selected text and extracts the most relevant information to produce a shorter and clearer version.

The generated summary maintains the academic tone of the original content while reducing unnecessary details. This functionality helps researchers quickly review large sections of text and identify the main contributions or findings of a document.

In experimental testing, the summarization module successfully condensed complex academic paragraphs into shorter summaries that preserved the key concepts and arguments. This demonstrates the effectiveness of transformer-based summarization models for academic text processing.

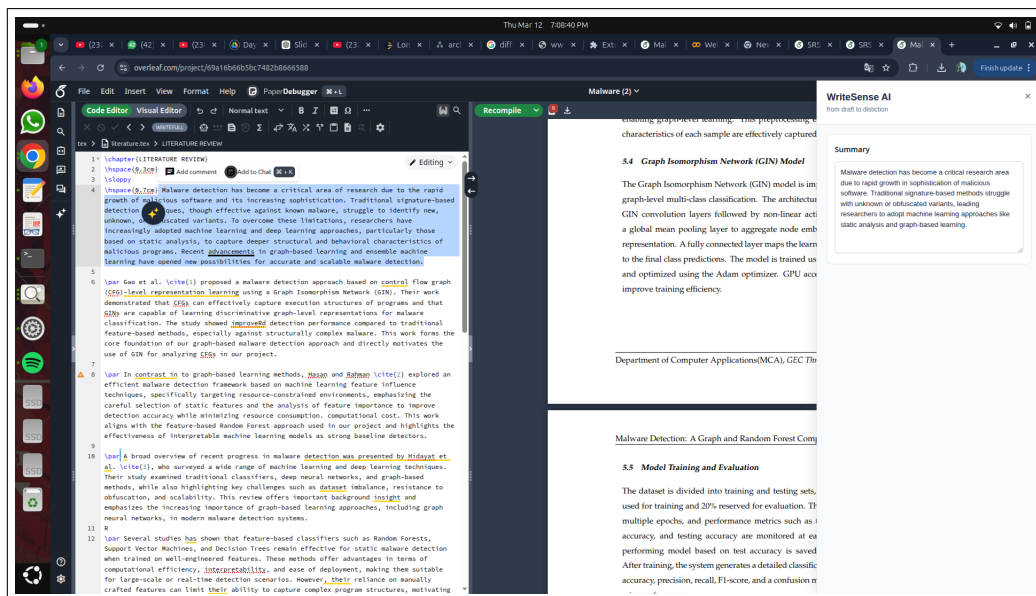


Fig. 7.4: Text Summarization Output

7.5 Multilingual Translation Results

Figure 7.5 shows the output of the Multilingual Translation Module. This module enables the automatic translation of selected academic text into multiple languages using the NLLB (No Language Left Behind) multilingual translation model. The translation functionality is designed to assist researchers who need to communicate academic content across different languages while maintaining the meaning and clarity of the original text.

When the user selects a portion of text in the Overleaf editor and chooses a target language, the system sends the selected content along with the language parameter to the translation backend service. The backend then processes the request using the NLLB transformer-based translation model. This model is trained on a large multilingual dataset and is capable of translating text across a wide range of languages while preserving contextual meaning.

During experimentation, the system successfully translated academic English text into several target languages. The generated translations preserved the logical structure of the sentences and maintained the semantic meaning of the original content. The translated output was observed to retain grammatical correctness and readability, which is essential for academic communication. This demonstrates that the translation model is capable of handling complex academic sentences and technical terminology.

Another important aspect of the translation module is its ability to preserve LaTeX syntax and document structure during translation. Academic documents often contain LaTeX commands, mathematical expressions, citations, and formatting instructions that should not be modified during language processing. The system therefore protects LaTeX commands before sending the text to the translation model and restores them after translation. This ensures that the translated content remains compatible with the LaTeX compilation process.

The translation module supports multiple languages and provides near real-time translation results within the extension interface. The results are

displayed in the dock panel of the Chrome extension, allowing users to review the translated text without leaving the writing environment. This integration significantly improves the efficiency of multilingual writing tasks and reduces the need for external translation tools.

This functionality is particularly beneficial for international researchers, postgraduate students, and academic authors who collaborate across different linguistic backgrounds. Researchers can quickly translate sections of their work to share ideas with collaborators, understand foreign-language research material, or prepare multilingual documentation.

Experimental testing confirmed that the translation system produces accurate and context-aware translations for academic content. The translated sentences retained their intended meaning while maintaining grammatical correctness and academic tone. These results demonstrate that the multilingual translation module effectively supports cross-language academic communication within the WriteSense AI writing environment. Furthermore, the system exhibited strong consistency across different subject domains, indicating its adaptability to varied academic contexts.

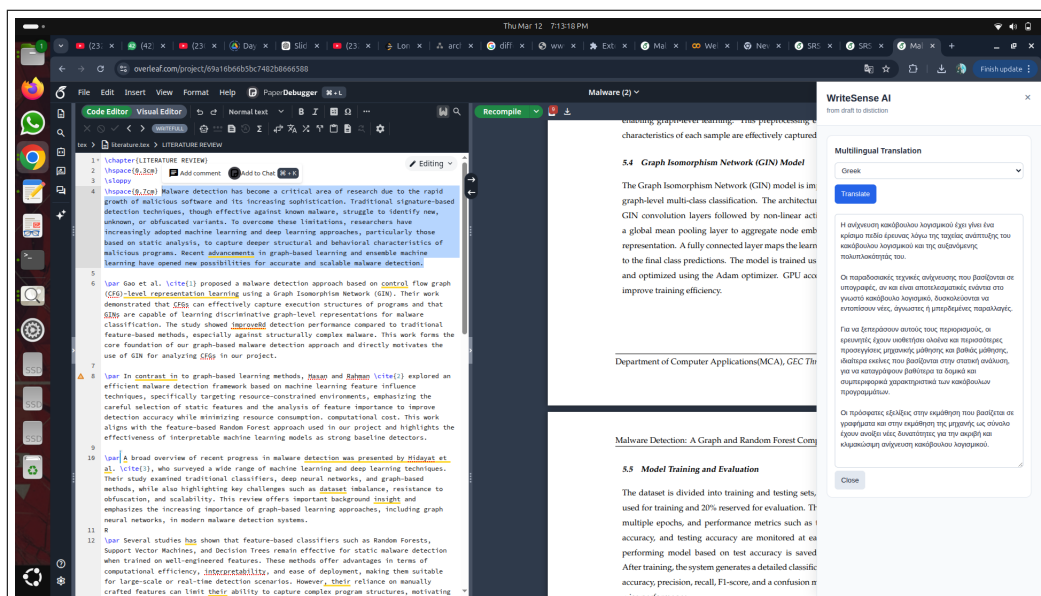


Fig. 7.5: Multilingual Translation Output

7.6 Research Paper Identification Results

Figure 7.6 illustrates the output of the Research Paper Identification Module. This module assists users in identifying relevant academic literature related to the selected text within the LaTeX document. The system first analyzes the selected academic paragraph and extracts important technical keywords that represent the main research concepts discussed in the text.

Keyword extraction is performed using AI-based text analysis techniques that identify significant terms and domain-specific expressions from the selected content. These keywords typically represent the core research topic, methodology, or technological domain of the document. For example, when testing the system with research paragraphs related to artificial intelligence and natural language processing, the system successfully extracted keywords such as machine learning, natural language processing, and language models. These extracted keywords form the basis for retrieving relevant scholarly publications.

After keyword extraction, the system sends a query request to the CrossRef scholarly database. CrossRef is a widely used academic metadata service that provides access to millions of research publications from journals, conference proceedings, and academic publishers. The backend service constructs search queries using the extracted keywords and retrieves a list of related research papers.

The retrieved results include important bibliographic information such as paper titles, author names, publication year, and DOI (Digital Object Identifier) links. These details allow users to easily identify and access relevant research articles that are related to the selected text. By presenting this information directly within the writing environment, the system helps authors quickly explore existing literature without leaving the editor.

To improve the relevance and usefulness of the results, the system applies a ranking mechanism based on keyword similarity and publication recency. Papers containing a higher number of matching keywords are

assigned greater relevance scores. In addition, recently published papers are prioritized to ensure that users receive up-to-date research information. This ranking strategy helps ensure that the most relevant and recent publications appear at the top of the results list.

Another important feature of the research paper identification module is the automatic generation of IEEE-style citations. Once relevant papers are retrieved from the CrossRef database, the system formats the bibliographic information into standard IEEE citation format. This allows users to quickly copy and integrate the citations into their LaTeX documents without manually formatting references.

During experimental evaluation, the module was tested using several academic text samples from different domains. In most cases, the system successfully identified the primary research domain and retrieved papers that were closely related to the topic of the selected text. The retrieved results were found to be contextually relevant and useful for literature exploration.

The experimental validation confirms that the research module accurately identifies research domains, extracts meaningful keywords, and retrieves contextually relevant academic publications. By integrating research paper discovery directly within the writing environment, the system helps authors efficiently explore related work while continuing their writing process.

In addition to retrieving relevant publications, the research paper identification module significantly improves the literature review process for academic authors. Traditionally, researchers must manually search for related work using external academic search engines such as Google Scholar or IEEE Xplore. This process often interrupts the writing workflow and requires switching between multiple applications. By integrating literature discovery directly into the writing environment, the proposed system reduces the need for such interruptions and allows researchers to remain focused on their document preparation.

Another advantage of this module is its ability to provide contextual

research suggestions based on the actual content of the document rather than generic keyword searches. Since the keywords are extracted directly from the selected paragraph, the retrieved research papers are more closely aligned with the topic being discussed. This context-aware retrieval mechanism improves the relevance of suggested literature and assists authors in identifying important references that may strengthen their academic arguments.

During experimental testing, several paragraphs from different research domains were used as input to evaluate the performance of the module. The system consistently extracted meaningful keywords and successfully retrieved research papers related to the underlying topic. The ranking mechanism also helped highlight recent and highly relevant publications, enabling users to quickly review important research contributions in the selected domain.

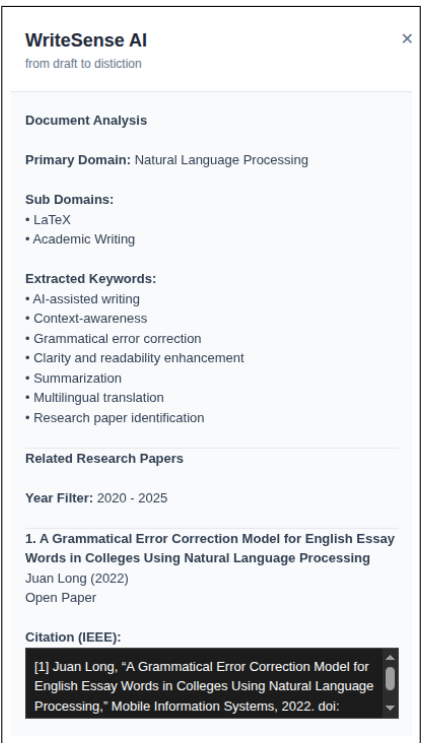


Fig. 7.6: Research Paper Identification Module Output

The experimental results demonstrate that the proposed WriteSense AI

system successfully integrates multiple academic writing assistance tools within a unified environment. Each module performs its respective task efficiently while maintaining the overall responsiveness of the system.

The hybrid grammar correction module effectively combines rule-based detection with AI-based structural improvement to enhance grammatical accuracy and sentence clarity. The clarity analysis module provides meaningful insights into readability and structural quality, enabling authors to improve the academic tone and organization of their writing.

The summarization module assists users in quickly understanding large sections of text, while the multilingual translation module enables access to academic content across language barriers. Furthermore, the research paper identification module supports literature discovery directly within the writing environment.

Overall, the integrated system significantly improves academic writing efficiency, reduces reliance on external tools, and ensures that users maintain full control over document modifications. The results confirm that WriteSense AI provides a practical and effective solution for enhancing the academic writing process within LaTeX-based environments.

CHAPTER 8

CONCLUSION

The WriteSense AI system provides a simple and practical solution for improving academic writing. It combines useful features like grammar correction, clarity checking, summarization, translation, and basic research support into a single platform. Instead of treating these as separate tools, everything is brought together within a LaTeX-based writing environment, making the overall writing process smoother and more convenient.

The platform allows users to work on their documents while getting suggestions that help improve readability, correctness, and structure. The summarization feature helps in quickly understanding content, while translation makes it easier to work with different languages. The research support feature also helps users find relevant papers, which makes writing more informed and efficient.

All these features are available through a simple browser interface, so users don't need any complicated setup or advanced technical knowledge. The system is designed to be lightweight but still useful, making it suitable for students as well as researchers.

Overall, WriteSense AI works like a helpful writing assistant that not only fixes mistakes but also helps improve the overall quality of writing. By bringing different AI-based features into one place, it makes the writing process easier and helps users create clearer and better-structured documents.

CHAPTER 9

FUTURE ENHANCEMENT

The proposed system offers several opportunities for future enhancements to further improve its functionality and usability. One potential improvement is the integration of a real-time writing assistant capable of providing instant suggestions as the user composes text. The summarization module can be extended to support multiple types of summaries, such as abstractive and extractive forms, to better suit different user needs. Additionally, the system can be expanded to support multiple languages, enabling users to write and analyze content in diverse linguistic contexts.

Furthermore, the inclusion of a plagiarism detection module would enhance the system's academic integrity features. Another possible enhancement is the development of a personalized feedback mechanism that adapts to individual writing styles, thereby offering more tailored suggestions. The user interface can also be improved by incorporating more interactive elements, such as visually highlighting grammatical errors and suggested corrections for better clarity. Finally, the integration of cloud storage capabilities would allow users to save, access, and manage their work efficiently across different devices.

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